EXP. 6.Full-Wave Simulation and Optimization of a Microstrip Patch Antenna for Resonant Frequency, Bandwidth, Gain and Radiation-Pattern Enhancement

Aim

To design and simulate a rectangular microstrip patch antenna using MATLAB and analyze its resonant frequency, return loss, VSWR, bandwidth, radiation pattern, gain, directivity, and efficiency.

Objectives

1. To calculate the dimensions of a rectangular microstrip patch antenna for a specified operating frequency.
2. To analyze the variation of S_{11}, VSWR, and input impedance with frequency.
3. To determine the resonant frequency, bandwidth, radiation pattern, gain, directivity, and efficiency.
4. To optimize the patch length for improved resonant-frequency performance.

Program:

```
clc;
clear;
close all;

%% MICROSTRIP PATCH ANTENNA
% Design frequency = 2.4 GHz

f0 = 2.4e9;
c = 3e8;

er = 4.4;
h = 1.6e-3;

%% 1. CALCULATE PATCH DIMENSIONS

W = c/(2*f0)*sqrt(2/(er+1));

eeff = (er+1)/2 + ...
      ((er-1)/2)*(1+12*h/W)^(-0.5);

dL = 0.412*h*((eeff+0.3)*(W/h+0.264))/ ...
    ((eeff-0.258)*(W/h+0.8));

L = c/(2*f0*sqrt(eeff))-2*dL;

fprintf('PATCH WIDTH  = %.2f mm\n',W*1000);
fprintf('PATCH LENGTH = %.2f mm\n',L*1000);

%% 2. FREQUENCY RANGE
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f = linspace(1.8e9,3.0e9,1000);

%% 3. APPROXIMATE RESONANT FREQUENCY

fr = c/(2*(L+2*dL)*sqrt(eeff));

fprintf('RESONANT FREQUENCY = %.3f GHz\n',fr/1e9);

%% 4. S11 / RETURN LOSS

% Approximate antenna impedance model

Rin = 50;

Q = 35;

X = 50*(f/fr - fr./f);

Z = Rin + 1j*X;

Gamma = (Z-50)./(Z+50);

S11 = 20*log10(abs(Gamma));

%% 5. PLOT S11

figure;

plot(f/1e9,S11,'LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
ylabel('S_{11} (dB)');
title('Return Loss (S_{11}) of Microstrip Patch Antenna');

hold on;

yline(-10,'--');

xline(fr/1e9,'--');

legend('S_{11}','-10 dB limit','Resonant Frequency');

%% 6. VSWR

VSWR = (1+abs(Gamma))./(1-abs(Gamma));

figure;

plot(f/1e9,VSWR,'LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
```

```
ylabel('VSWR');
title('VSWR versus Frequency');

hold on;

yline(2,'--');

legend('VSWR','VSWR = 2');

%% 7. INPUT IMPEDANCE

figure;

plot(f/1e9,real(Z),'LineWidth',2);

hold on;

plot(f/1e9,imag(Z),'LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
ylabel('Impedance (Ohm)');

title('Input Impedance versus Frequency');

legend('Resistance','Reactance');

%% 8. -10 dB BANDWIDTH

BWindex = find(S11 <= -10);

if ~isempty(BWindex)

   fL = f(BWindex(1));
   fH = f(BWindex(end));

   BW = fH-fL;

   fprintf('\nLOWER FREQUENCY = %.3f GHz\n',fL/1e9);
   fprintf('UPPER FREQUENCY = %.3f GHz\n',fH/1e9);
   fprintf('BANDWIDTH = %.2f MHz\n',BW/1e6);

   fprintf('FRACTIONAL BANDWIDTH = %.2f %%\n', ...
      (BW/fr)*100);

else

   fprintf('No -10 dB bandwidth found.\n');

end

%% 9. RADIATION PATTERN

theta = linspace(0,2*pi,720);
```

```
% Approximate broadside patch radiation pattern

E = abs(cos(theta));

E = E/max(E);

figure;

polarplot(theta,E,'LineWidth',2);

title('Radiation Pattern of Microstrip Patch Antenna');

%% 10. 3D-LIKE RADIATION PATTERN

phi = linspace(0,2*pi,180);
theta3 = linspace(0,pi,180);

[PHI,THETA] = meshgrid(phi,theta3);

R = abs(sin(THETA).*cos(THETA));

R = R/max(R(:));

X3 = R.*sin(THETA).*cos(PHI);
Y3 = R.*sin(THETA).*sin(PHI);
Z3 = R.*cos(THETA);

figure;

surf(X3,Y3,Z3,R);

shading interp;

axis equal;

xlabel('X');
ylabel('Y');
zlabel('Z');

title('3D Radiation Pattern');

%% 11. GAIN AND DIRECTIVITY - APPROXIMATE

D = 6.5;              % Approximate directivity in dBi
efficiency = 0.85;    % Approximate efficiency

Gain = D + 10*log10(efficiency);

fprintf('\nDIRECTIVITY = %.2f dBi\n',D);
fprintf('RADIATION EFFICIENCY = %.2f %%\n',efficiency*100);
fprintf('GAIN = %.2f dBi\n',Gain);

%% 12. PATCH LENGTH OPTIMIZATION
```

```
lengths = L*(0.90:0.01:1.10);

resFreq = zeros(size(lengths));

for k = 1:length(lengths)

    Ltest = lengths(k);

    dLtest = 0.412*h*((eeff+0.3)*(W/h+0.264))/ ...
          ((eeff-0.258)*(W/h+0.8));

    resFreq(k) = c/(2*(Ltest+2*dLtest)*sqrt(eeff));

end

%% 13. OPTIMIZATION GRAPH

figure;

plot(lengths*1000,resFreq/1e9,'LineWidth',2);

grid on;

xlabel('Patch Length (mm)');
ylabel('Resonant Frequency (GHz)');

title('Patch Length Optimization');

yline(f0/1e9,'--');

%% 14. FIND BEST LENGTH

[~,index] = min(abs(resFreq-f0));

optimalLength = lengths(index);

fprintf('\nOPTIMIZED PATCH LENGTH = %.2f mm\n', ...
    optimalLength*1000);

fprintf('OPTIMIZED RESONANT FREQUENCY = %.3f GHz\n', ...
    resFreq(index)/1e9);

%% 15. OPTIMIZED S11

frOpt = resFreq(index);

Xopt = 50*(f/frOpt - frOpt./f);

Zopt = 50 + 1j*Xopt;

GammaOpt = (Zopt-50)./(Zopt+50);

S11opt = 20*log10(abs(GammaOpt));

figure;
```

```
plot(f/1e9,S11,'LineWidth',2);

hold on;

plot(f/1e9,S11opt,'--','LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
ylabel('S_{11} (dB)');

title('Original and Optimized Return Loss');

legend('Original','Optimized');

yline(-10,'--');

%% FINAL RESULTS

fprintf('\n========== FINAL RESULTS ==========\n');

fprintf('Operating Frequency = %.2f GHz\n',f0/1e9);

fprintf('Patch Width = %.2f mm\n',W*1000);

fprintf('Patch Length = %.2f mm\n',L*1000);

fprintf('Resonant Frequency = %.3f GHz\n',fr/1e9);

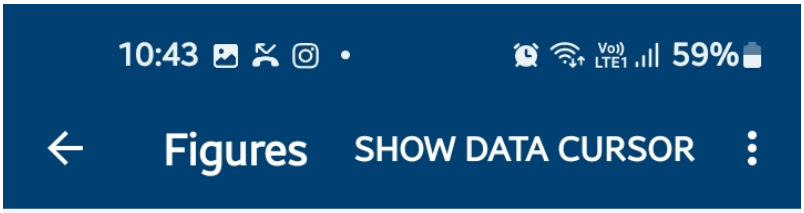
fprintf('Minimum S11 = %.2f dB\n',min(S11));

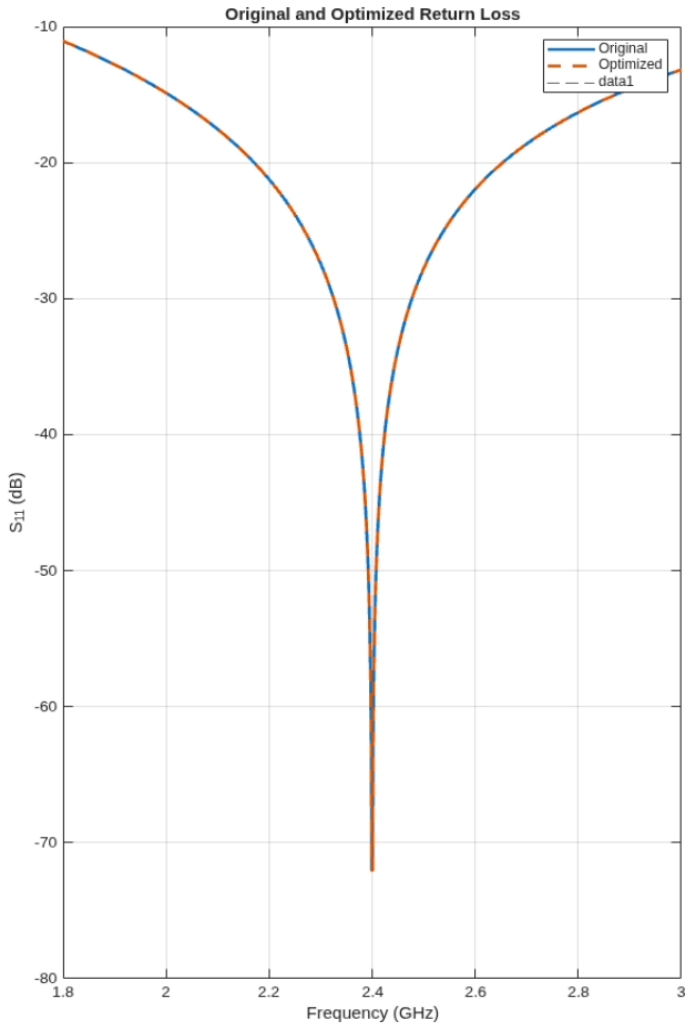
fprintf('Directivity = %.2f dBi\n',D);

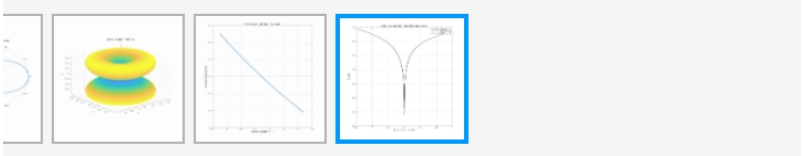
fprintf('Gain = %.2f dBi\n',Gain);

fprintf('Efficiency = %.2f %%\n',efficiency*100);
```

Output:


10:43
59%
Figures
SHOW DATA CURSOR
Figure 7: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...


Original and Optimized Return Loss
Original
Optimized
data1
S11 (dB)
Frequency (GHz)
-10
-20
-30
-40
-50
-60
-70
-80
1.8
2
2.2
2.4
2.6
2.8
3



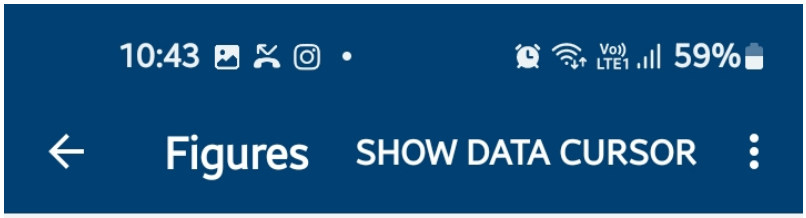
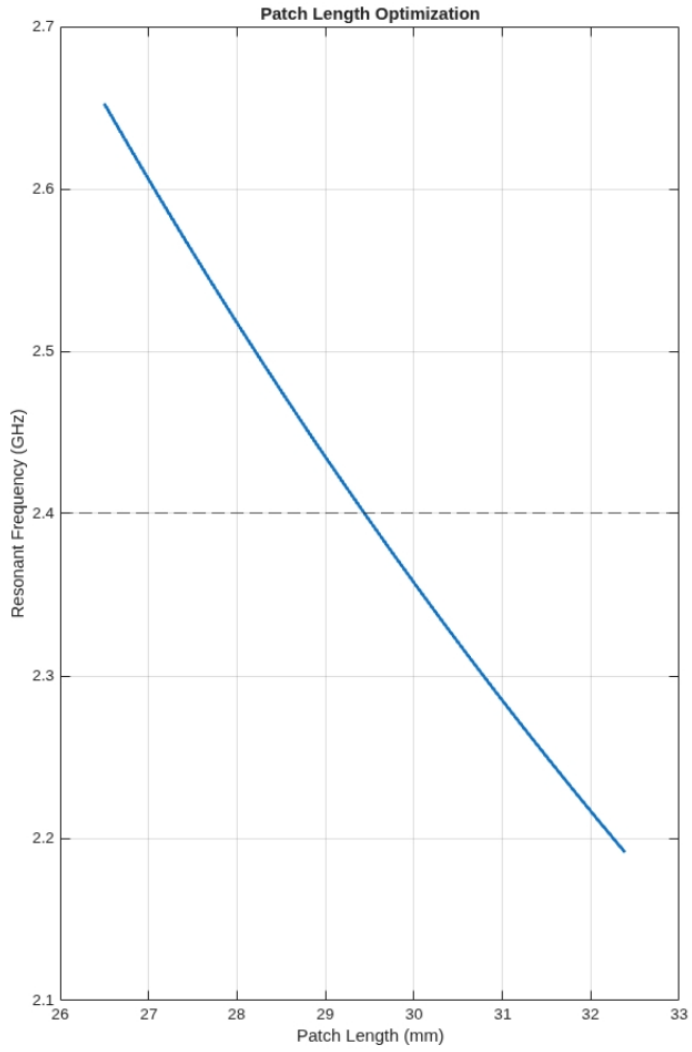
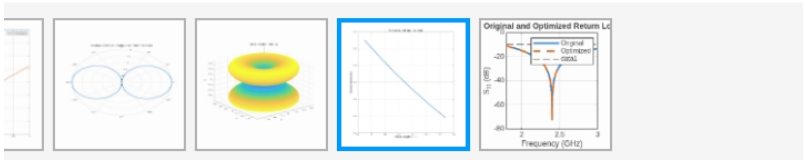


Figure 6: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





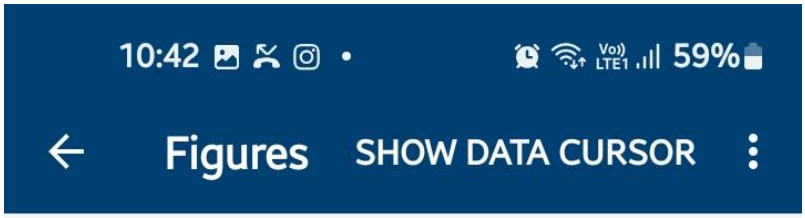
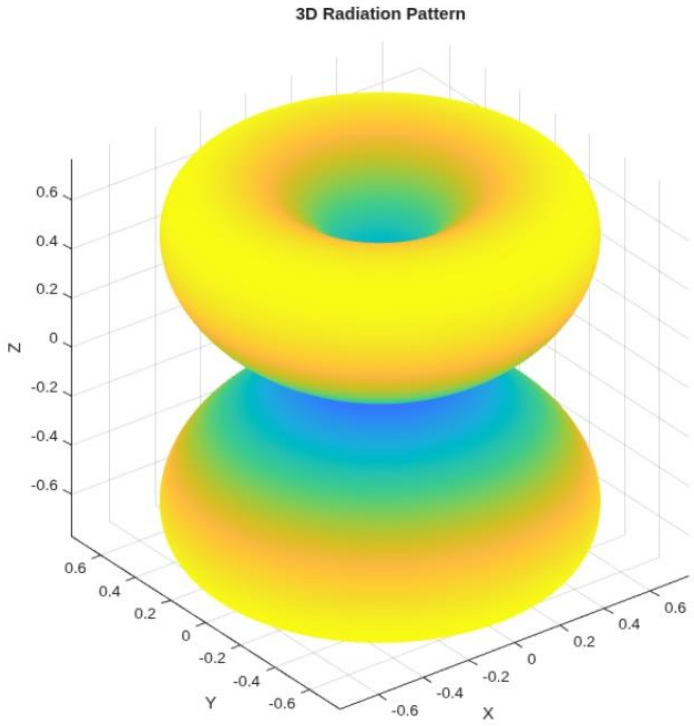
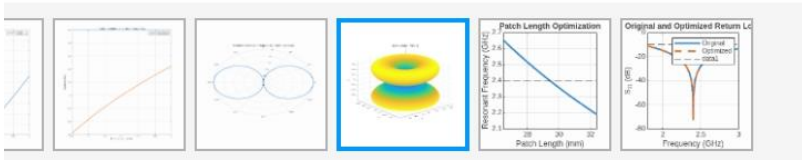


Figure 5: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





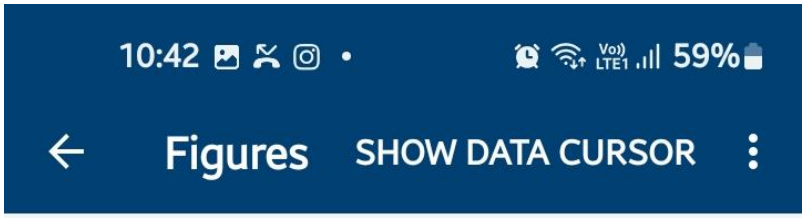
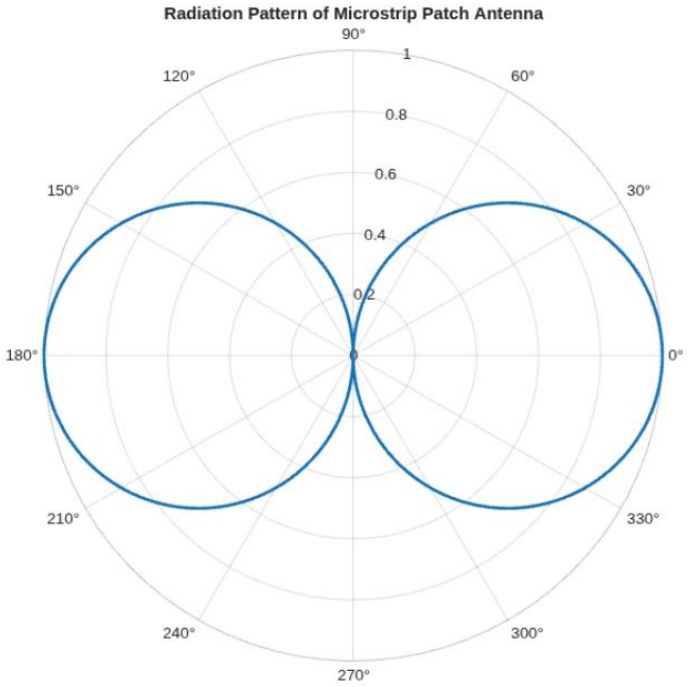
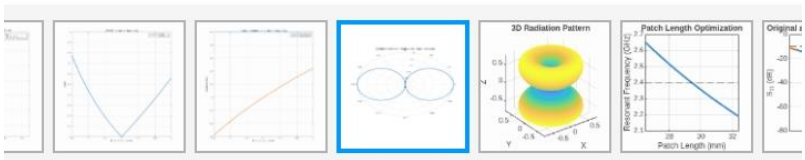


Figure 4: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





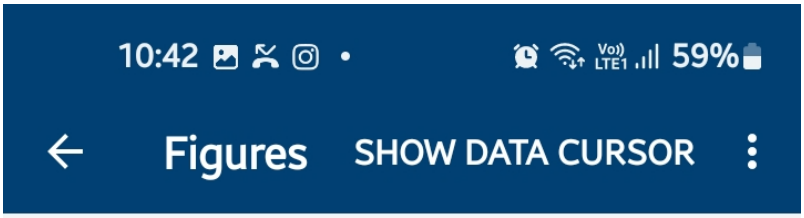
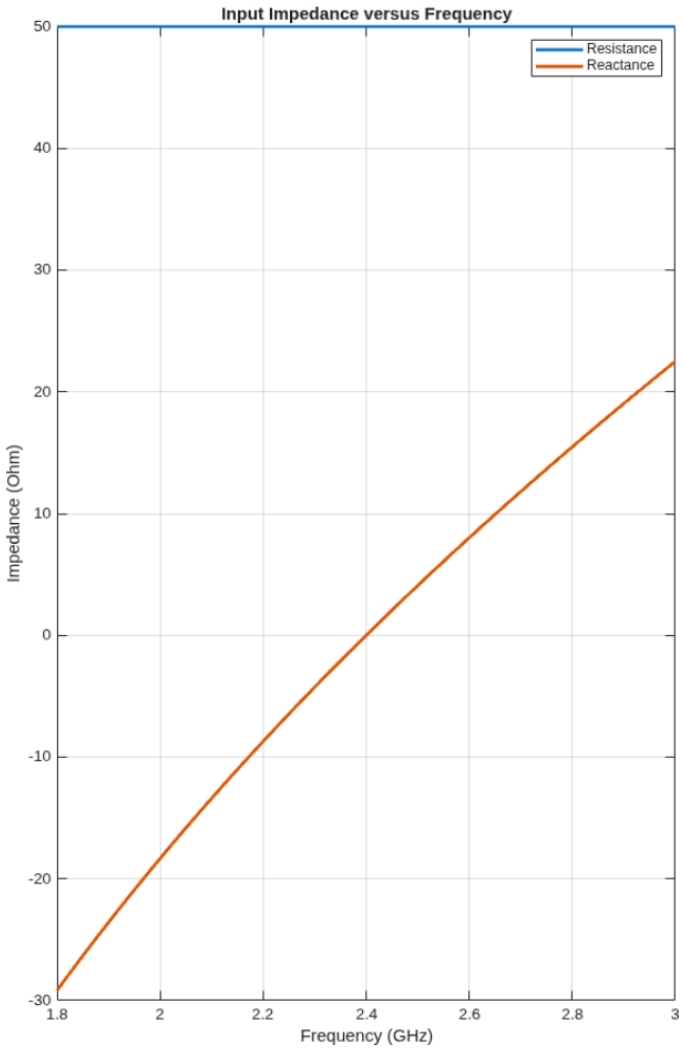
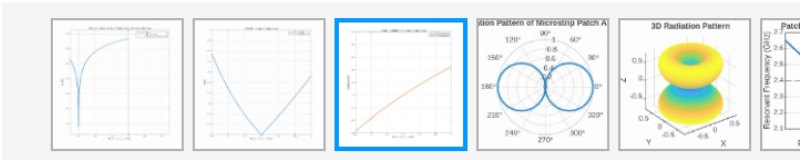


Figure 3: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





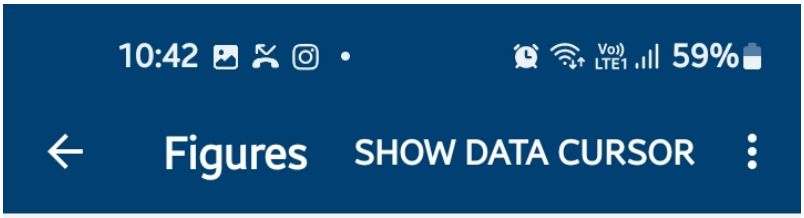
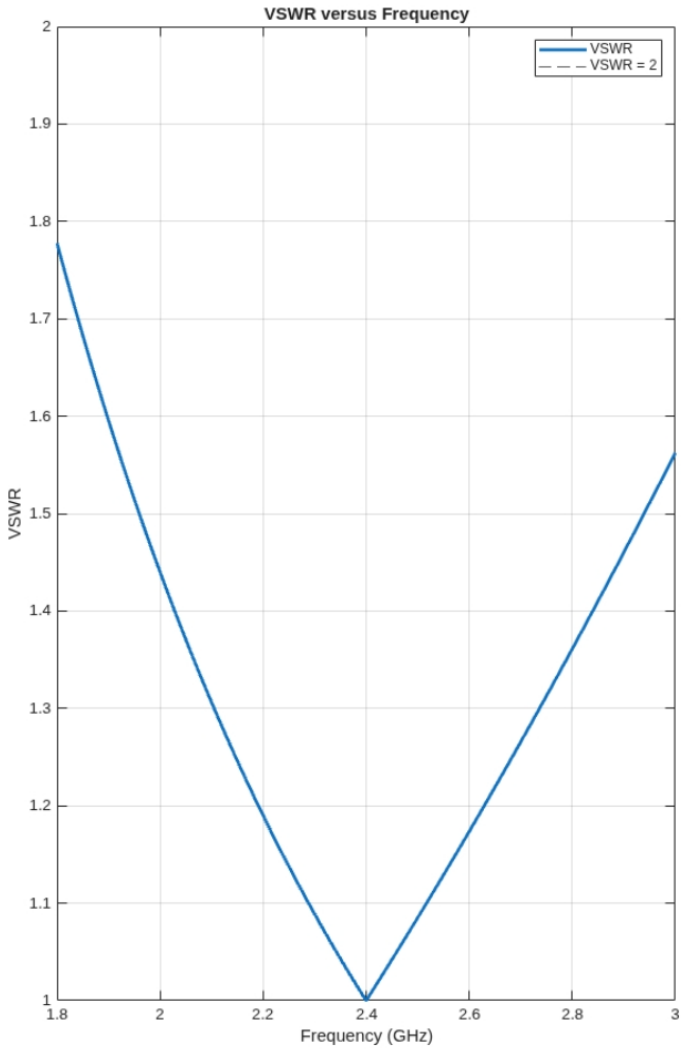
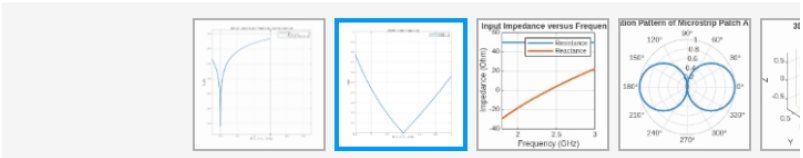


Figure 2: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





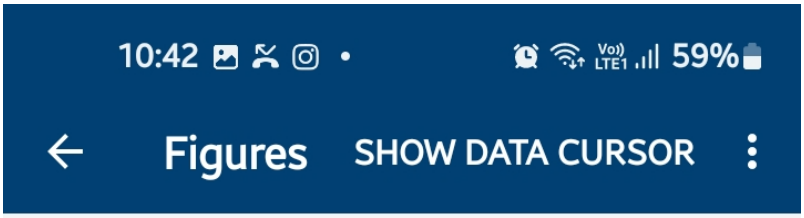
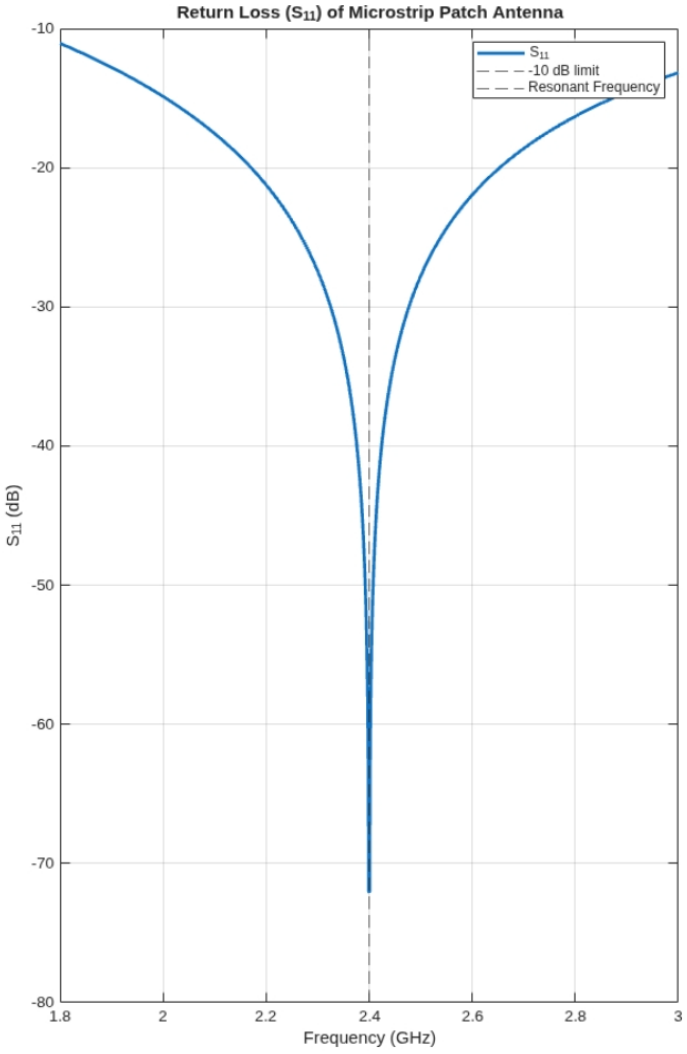
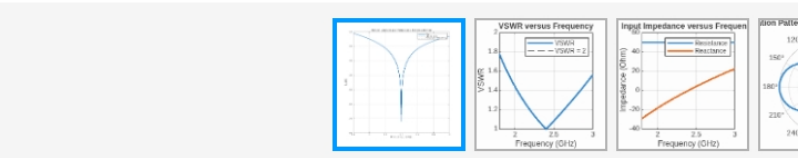


Figure 1: clc; clear; close all; %% MICROSTRIP PATCH ANTENN...





Reflection

The MATLAB simulation helped me understand the relationship between the **patch dimensions and antenna performance**. The S_{11} and VSWR plots were used to identify the resonant region and impedance-matching behavior. The radiation pattern showed the directional radiation characteristics of the patch antenna. Optimization of the patch length demonstrated how small geometrical changes can shift the resonant frequency toward the desired operating frequency.

Result

The rectangular microstrip patch antenna was successfully designed and simulated for an operating frequency of **2.4 GHz**. The patch dimensions were calculated as approximately **38.04 mm width** and **29.44 mm length**. The S_{11}, VSWR, impedance, radiation pattern, and optimization characteristics were obtained using MATLAB. The simulation demonstrated that adjusting the patch length can improve the antenna's resonant-frequency performance.